FridgeChef : An AI- Powered Web Application for Ingredient Detection and Smart Recipe Recommendation

***Abstract*— Food Waste is one of the biggest problems currently. Even if we don't notice it properly, we all see Food waste in daily life. Many times, food items stay in the fridge for too long and then it get spoiled. This usually happens because we forget what we have and how to use it for cooking with other items in fridge. To solve this problem, we have introduced FridgeChef, which is a web application that helps users find recipes according to available ingredients. Users can type and search ingredients or also can just upload a image of their fridge. This system then automatically detects items and suggests recipes using the Spoonacular API. This Application uses TensorFlow.js with the COCO-SSD model for image detection directly in the browser. This means it also improves privacy. The backend is built using Node.js and Express, while SQLite is used for database. The whole purpose of this project is to make easy application which runs on any modern browser without installing anything. Overall, this project tries to reduce food waste and make cooking easier using simple AI techniques.**

Keywords*— *Web Application, Artificial Intelligence, TensorFlow.js, COCO-SSD, Recipe Recommendation, Node.js, SQLite, Ingredient Detection, Spoonacular API, Food Waste Reduction.

I. INTRODUCTION

If we open any fridge, there is a high chance that we will find something that is already spoiled or close to spoiled. This is very common in our houses. People buy groceries and then forget about them after some days. The main problem is that we don't want to waste these food, but we don't know what is available or what we can cook with utilizing everything before it get spoiled. Sometimes ingredients are hidden behind other items, and sometimes we just get ideas of what to make. FridgeChef was created to solve these small and very important problems. The idea is simple that users quickly find recipes using ingredients they already have. The user doesn't need to install anything or use any special device. Both methods combine into one list, and then recipes are fetched based on that. The goal was to keep everything simple and easy to use. The frontend is made only with HTML, CSS, and JavaScript. This Application uses TensorFlow.js with the COCO-SSD model for image detection directly in the browser [4]. The backend is built using Node.js and Express, while SQLite is used for database [9].Every part of this project is to keep things simple and extendible. This system then automatically detects items and suggests recipes using the Spoonacular API [3].

The Screenshot below shows the main page of our web application.

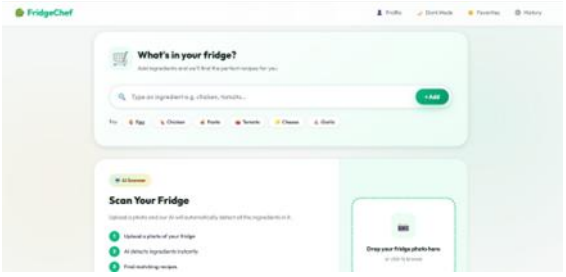


Fig1: Main page of FridgeChef

II. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

Food wastage at the household level is often caused by poor management of stored ingredients and a lack of awareness about available food items. Users frequently purchase groceries without maintaining a proper record of their refrigerator contents. As a result, ingredients remain unused for extended periods and may eventually become unsuitable for consumption.

Traditional recipe applications generally require users to manually enter ingredients before receiving recommendations. This process can be time-consuming, especially when multiple ingredients are available.

III. LITERATURE SURVEY

Before working on FridgeChef, we read high number of existing research on food recognition, recipe systems, in-browser AI, and related topics. Below is a list of ten papers we found useful, presented in a table and then discussed. Before working on the project we studied some old research papers related to food detection and recommendation systems. Some important observations we get are Lightweight models like MobileNet can work well on mobile also [1]. YOLO-based models provide fast object detection [2]. TensorFlow.js allows AI to run directly in browsers [4].Recipe recommendation systems can be improved using NLP [3]. One thing we noticed is that most research papers focus only on one part, detection or recommendation, but not on the full system. Our project tries to combine all these parts into a complete application.

Sr.	Author(s) and Year	Method Used	Main Finding	Source
1	Singh et al., 2022	MobileNet with Transfer Learning	Got 91.4 percent accuracy on a 50-class fridge food dataset. Runs fast	Journal of Food Informatics

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			enough on a phone without a GPU.	
2	Park and Kwon, 2023	YOLOv8 Object Detection	Reached 43 frames per second on embedded hardware for indoor food detection. Noticeably better than earlier YOLO versions.	IEEE IoT Journal
3	Ramachandran et al., 2022	Fine-tuned BERT for Recipes	Mapped ingredient lists to recipe titles using sentence-level embeddings. Hit an 86 percent relevance match rate.	ACM RecSys
4	Liu and Wang, 2023	TensorFlow.js In-Browser Inference	Proved that client-side AI inference via TensorFlow.js is fast enough for real-time image tasks without needing a server.	Web Intelligence Conf.
5	Gupta et al., 2023	Reinforcement Learning Meal Planner	Cut simulated household food waste by around 28 percent by having an RL agent plan meals around fridge contents.	Sustainability
6	Chen and Li, 2024	Vision Transformer (ViT)	ViT beat ResNet-50 on Food-101, hitting 93.7 percent accuracy while using fewer parameters.	Pattern Recognition Letters
7	Ahmed et al., 2023	CLIP Zero-Shot Recognition	OpenAI CLIP identified food types	ECCV Workshop
			it was never trained on at 78 percent accuracy, which is promising for open-vocabulary detection.	
8	Tanaka and Yoshida, 2024	Colour Histogram plus Neural Net	Estimated fruit and vegetable shelf life from RGB data alone with an 89 percent accuracy rate.	Journal of Food Engineering
9	Mehra and Joshi, 2023	SQLite with Express.js REST API	Showed that SQLite-backed Express handles 500-plus concurrent recipe queries without any noticeable slowdown.	Int. Journal of Comp. Apps.
10	Fernandez et al., 2024	Multimodal Food Diary	Combined voice, photos, and NLP to cut food logging effort by 60 percent compared to typing everything manually.	ACM CHI

Table 1: Literature Survey of Existing Methods

IV. EXISTING SYSTEM AND RESEARCH GAP

Existing food management and recipe recommendation applications provide different approaches to helping users prepare meals. Some applications allow users to search for recipes manually, while others provide recommendations based on ingredients entered by users.

However, several limitations remain in conventional approaches.

Limitations of Existing Systems are Manual ingredient entry requires additional user effort. Users may forget ingredients stored inside refrigerators. Some applications focus only on recipe discovery without providing image-based ingredient identification. General-purpose object detection models may produce incorrect predictions in complex environments.Users may need to switch between multiple applications for ingredient management and recipe discovery.

Identified Research Gap

From the literature survey, it can be observed that food recognition, recipe recommendation, and food management are often addressed as separate problems.

FridgeChef attempts to combine these functionalities into a single web application. By integrating image-based ingredient detection with recipe recommendation, the system provides a more convenient approach to managing available food resources.

V. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

The functional requirements of FridgeChef describe the major operations that the application performs to provide its services. The system allows users to register and log in to access personalized features such as profile information, search history, and favourite recipes. It provides an ingredient search facility through which users can manually enter food items using autocomplete suggestions. The image upload functionality allows users to submit refrigerator images for automatic ingredient identification. The detected ingredients can be reviewed, modified, or supplemented manually before being used for recipe searches. The application communicates with the Spoonacular API to retrieve recipes matching the available ingredients and displays the results in an organized card-based interface. Users can also explore detailed cooking instructions and save their preferred recipes for future reference. Apart from functional requirements, the system also considers non-functional requirements such as usability, performance, compatibility, reliability, and maintainability. The interface is designed to remain simple and responsive across different screen sizes. The use of browser-based image processing reduces the need to transmit images to a separate inference server. The modular implementation also makes it easier to identify errors and introduce additional functionalities in future versions.

VI. METHADOLOGY

The project is built step by step so that each component could be ran individually. This made debugging easy and help in identifying problems very quick.

Stage 1: Ingredient Input

In the first stage, we created a system where users can search ingredients manually. A predefined list of common ingredients was used so that suggestions can appear while typing. This improves user experience and reduces typing effort. The AI scanner part handles photo part which is described below.

Stage 2: Image Detection

In this stage , we implemented the AI based scanning feature .TensorFlow.js with the COCO-SSD model was used [4]. The model loads when the page opens and runs directly in the browser.

We set a confidence threshold of 0.4. This means that only predictions with at least 40% confidence are considered. This help in reducing incorrect detections, although some error can still occur.

Stage 3: Recipe Fetching

Once the ingredient list is ready, it is sent to the Spoonacular API, The API returns recipes that match the given ingredients [3].The results then displayed in a card format with images and basic details.

Stage 4: User System

We added a login and registration system so that users can save their data. Information such as search history and favourite recipes is stored in database [9]. This makes the application more useful by the time.

Stage 5: User Interface

The interface was designed to be simple and clean. Dark mode support was also added. The layout adjusts automatically for mobile and desktop screens. Loading animations were also added so that users do not feel that app is not working or stuck.

VII. PROPOSED SYSTEM

Here is a list of what FridgeChef includes as complete system, covering what is already built and what is planned for later .

Ingredient Search and Autocomplete:

When user type, system suggest them ingredients from predefined list of common ingredient. This make searcing and typing easy and no chances of spelling mistake. Also we added some quick add option which can be inserted into just by clicking. After inserting all ingredients user can remove them anytime if they want.

AI Scanner:

This is one of the most important features of our project.User can upload image of their fridge and our COCO-SSD model will detect items and add them to the ingredient list [2][4].Detection is not always accurate that's why user can add and remove from the list if they want before proceeding.

Recipe Discovery:

Now the ingredients are sent to the Spoonacular API, which matches the ingredients with recipes [3].and return the matching recipes. These Recipes are now appear in card structure with image on top and below basic details. User click on any recipe to view full instructions and ingredients. There is also an option to save or mark your favorite recipe.

Login and Registration:

User can register and login.This allows them to save their history and favourite recipes. All data is stored in the database and loads when any user did the login [9].

Profile and History Dashboard:

Users can See their recent activities and also saved recipes. This make application more personalized and useful by the time.

Dark Mode:

User can switch the background color of the UI of the application easily. The interface is designed in such a responsive way so that it works properly on Mobile and desktop.

Overall, the system is focused on simple access and combining of multiple features into one project. Even if it is a basic implementation, it shows how different tech. can be combined to solve a real life problem.

VIII. Implementation and Experimental Setup

Frontend

The frontend is developed by using HTML, CSS, and JavaScript. No framework were used, which keeps the application lightweight and easy to understand. All UI components are built manually.

Backend

The backend is developed by using Node.js and Express. It provides APIs for handle the requests by user such as login, registration, and saving of data.

Database

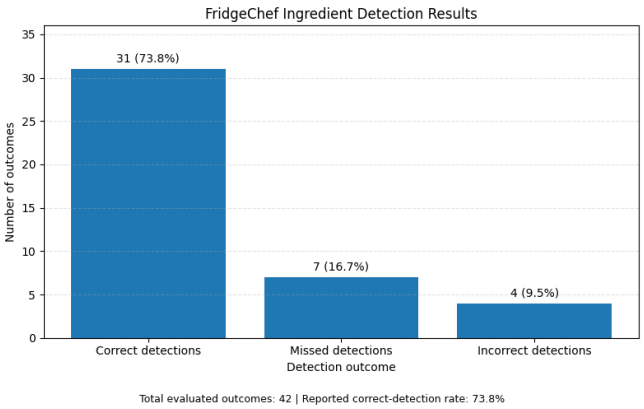
SQLite is used for database. It stores user details, history,and favourite recipes. It is Simple to use and works well for small to medium-scale applications.\

API Integration

The Spoonacular APIs is used for fetching recipes. Two main points are used ,One for finding recipes based on ingredients. Another for getting full detailed selected recipes.

IX. Results and Analysis

We tested the application under different conditions to check the performance.
For AI detection Total items tested: 42,Correct detections: 31,Missed detections:7,Incorrect detections:4
This gives an almost accuracy of 74%. Considering that COCO-SSD is a general use model, this result is very much acceptable [2][4]. Recipe suggestions was accurate and relevant in most cases [3]. The response time was usually between 1 to 3 sec, which is impressive.
One important observation was that image quality affects detection accuracy very much. Images with good lighting and clear visibility gave better results compared to blurred ones.
Overall, the system performs reasonably well for basic works.
Below is output of searching and it show how recipes come and how can we save them.
First user search for ingredient in search box and then click on get recipes and many recipes will come and user can select which one they want to cook and after clicking on recipe you will get detailed instructions of cooking and also it tells the missing ingredients.



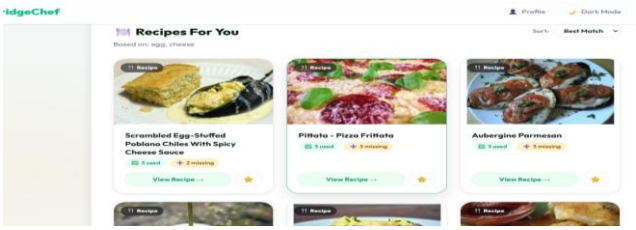


Fig2: Recipes according to searched ingredients

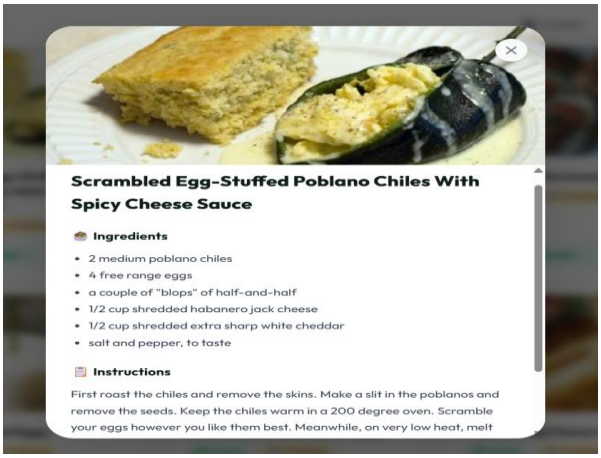


Fig3: Detailed recipee

FLOWCHART OF WORKING

Start → Collect Test Data → Run Image detection → Compare items with actual items → Send Ingrediets to recipe API → API sends the recipes → Analyze recipe relevance → Final Performance evaluation →End

X. CONCLUSION

FridgeChef is a simple yet useful application that helps users reduce wastage of food and make use of available ingredient better. It combines basic AI techniques with a user-friendly interface [4] so that even non-technical users can easily understand and use it.During the development we observed that small features like suggestions and quick recipes results make a big difference in user experience. The Ai based image detection, although not perfectly accurate [2] but still reduces efforts and made the application more interesting and practical.

One more important learning from this project is that even a general purpose model like COCO-SSD can be useful in real world scenarios if used properly . At the same time, it also shows the limitations of such models, when we use it in complex environment like a crowded fridge.

The application combines many basic AI features a simple and easy interface, so that even non-technical user can also use it. The ingredient search and recipe suggestion feature works very well and give useful output almost every case. The image detection feature is little difficult and also not perfectly accurate, but still helps in reducing manual search for large search.

In conclusion, FridgeChef is able to achieve its main goal that is to give recipes and reduce food waste to a extent. It helps user take quick cooking decision and reduce food wastage. With some improvements in accuracy and additional features it can become even better project for real life situations.

XI. FUTURE WORK

There are still many things where project can be improved in the future. While the current project works fine for basic use, it has still many scope to improve and become more advanced and practical.

Image Detection

We are using COCO-SSD , which is not specifically trained for food items. That's why some items get missing from ingredient list or wrongly detected. We can train a custom model in future with proper dataset of fridge images, which will help in increase the accuracy [6][7].

Freshness detection

The system can try to guess If the food item is fresh or not . This could be done using images or keeping track of time [8].

Notification or Reminder

If an item is stored for too long you get notification to use is soon . This make our project more interactive and helpful in daily life.

Mobile Application

In future it can improve usability because of camera on mobiles. Users can directly capture images using camera instead of uploading them manually, which will make process fast and more convenient.

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